

1. Driver Authentication & Profile Management

The Driver App needs to authenticate the courier against the existing vendor database and establish their current working state.

- **Features:**
 - Driver login using vendor-created credentials.
 - Session token persistence so the driver stays logged in during their shift.
 - Online / Offline availability status toggle.
 - Push notification token registration upon login.
- **Backend & APIs:**
 - POST /api/driver/auth/login — Authenticates driver credentials and returns a secure JWT token along with vendor/vehicle profile info.
 - POST /api/driver/auth/push-token — Saves or updates the driver's active push notification token (device_token) in the database.
 - PATCH /api/driver/status — Updates availability status (online / offline) in the drivers table.

2. Assigned Route & Order Fetching

The Driver App fetches the optimized route and assigned order sequence generated by the vendor.

- **Features:**
 - View today's assigned route overview (total stops, estimated duration, total distance).
 - View sequential list of stops with customer details, delivery time windows, addresses, and special notes.
 - Display route stops and order pins on the app map view.
- **Backend & APIs:**
 - GET /api/driver/routes/today — Fetches today's active route for the logged-in driver from the routes table.
 - GET /api/driver/routes/:route_id/deliveries — Fetches the sequenced list of stops, delivery addresses, customer contacts, and coordinates from the deliveries and orders tables.

3. Real-Time Delivery Status & Progress Sync

As the driver progresses through their shift, status changes made in the mobile app immediately update the Vendor Portal's live monitoring screen.

- **Features:**
 - Update order stage sequentially: *Arrived*, *Completed*, *Failed*, or *Delayed*.
 - Attach failure or delay reasons when an order cannot be completed.
 - Auto-timestamp each status transition.
- **Backend & APIs:**
 - PATCH /api/driver/deliveries/:delivery_id/status — Updates the stop status in the deliveries table and records automatic timestamps (arrived_at, delivered_at).
 - **Database WebSockets / Realtime Trigger:** Listens to changes on the deliveries table

and pushes live status updates directly to the vendor's monitoring dashboard.

4. Proof of Delivery Upload

When a package is handed over, evidence captured on the mobile app is attached directly to the vendor's matching order record.

- **Features:**
 - Capture customer signature or take delivery photos.
 - Upload compressed media files to cloud storage.
 - View uploaded proof links inside the Vendor Portal's delivery history.
- **Backend & APIs:**
 - `POST /api/driver/deliveries/:delivery_id/proof` — Uploads photo or signature binary to Supabase Storage (proofs bucket) and writes the resulting public URL to `deliveries.proof_url`.

5. Live GPS Location Streaming

The Driver App continuously streams live coordinates to display moving driver markers on the Vendor Portal map.

- **Features:**
 - Periodic background location streaming while on an active route.
 - Auto-pause tracking when the driver toggles to offline or completes the route.
- **Backend & APIs:**
 - `POST /api/driver/location` — Receives periodic latitude and longitude coordinates from the app and updates `drivers.current_location` or logs to `driver_locations`.
 - **WebSocket Channel (drivers:location):** Broadcasts coordinate changes instantly to the vendor's live map view without manual page reloads.

6. Dynamic Route Alerts & Push Notifications

The Vendor Portal pushes real-time notifications to the Driver App whenever changes occur on the server side.

- **Features:**
 - Instant alert when a new route is dispatched.
 - Notification if a vendor adds, reorders, or cancels a stop mid-shift.
- **Backend & APIs:**
 - **Webhook / Server Event Trigger:** When a vendor modifies a dispatched route in the portal, the backend uses Firebase Cloud Messaging (FCM) to push payload alerts directly to the driver's phone.
 - `GET /api/driver/notifications` — Fetches unread alert logs from the notifications table for display inside the driver app notification drawer.